Review of Linear Regression

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Outline

1. Causality and linear regression model.
2. Basic results about linear regression model.
3. Instrumental variables estimation.

Causality and linear regression model

We are usually interested in causality

- Consider the *causal effect* of schooling on earnings. Let s_i be a dummy variable of having a college degree; $s_i = 1$ if i has a diploma, and $= 0$ otherwise. Let y_i be earnings.
- We are interested in the *causal effect* of schooling on earnings: how does y_i change if s_i changes holding all other relevant factors fixed.
- *Causality* like the above is applied econometricians' main interest.

Potential outcomes and causal effect

- Let y_{1i} and y_{0i} be the potential earning of i if $s_i = 1$ and $s_i = 0$, respectively. The observed outcome is

$$y_i = y_{0i} + (y_{1i} - y_{0i})s_i.$$

- For college graduate $s_i = 1$, y_{0i} is i's earnings if i did not go to college holding all other factors, such as ability, fixed.
- The causal effect for i is then $\delta_i = y_{1i} - y_{0i}$.
- It is common to call causal effect (economists) as treatment effect (statisticians).

Difficulty in evaluating causal effects

- The difficulty in finding $\delta_i = y_{1i} - y_{0i}$ is that for any i, we can only observe either y_{1i} or y_{0i}, never both.
- Clearly, δ_i is individual specific. Not easy to summarize.
- Average treatment effect (ATE) $\mathrm{E}(y_{1i} - y_{0i})$.
- Average treatment effect on treated (ATT) $\mathrm{E}(y_{1i} - y_{0i} \mid s_i = 1)$.

Selection bias from naive group mean difference

- From data, we know group means $\mathrm{E}(y_i \mid s_i = 1)$ and $\mathrm{E}(y_i \mid s_i = 0)$.
- The naive group mean difference $\mathrm{E}(y_i \mid s_i = 1) - \mathrm{E}(y_i \mid s_i = 0)$ could be misleading in evaluating the causal effect of s.

$$\begin{aligned}\mathrm{E}(y_i \mid s_i = 1) - \mathrm{E}(y_i \mid s_i = 0) = \mathrm{E}(y_{1i} \mid s_i = 1) - \mathrm{E}(y_{0i} \mid s_i = 1) \\ + \mathrm{E}(y_{0i} \mid s_i = 1) - \mathrm{E}(y_{0i} \mid s_i = 0).\end{aligned}$$

- $\mathrm{E}(y_{0i} \mid s_i = 1) - \mathrm{E}(y_{0i} \mid s_i = 0)$ is "*selection bias*".
- When there is no selection bias, ATT equals

$$\mathrm{E}(y_{1i} - y_{0i} \mid s_i = 1) = \mathrm{E}(y_i \mid s_i = 1) - \mathrm{E}(y_i \mid s_i = 0).$$

Conditional independence assumption for eliminating selection bias

- *Conditional independence assumption* (CIA) says

$$(y_{1i}, y_{0i}) \perp\!\!\!\perp s_i \mid c_i,$$

  where c_i is a vector of observable control variables.
- CIA implies that $\mathrm{E}(y_{0i} \mid s_i = 1, c_i) = \mathrm{E}(y_{0i} \mid s_i = 0, c_i) = \mathrm{E}(y_{0i} \mid c_i)$. No selection bias.
- Rephrase CIA: consider two individuals i and j with $s_i = 1$ and $s_j = 0$. If $c_i = c_j$, CIA says (y_{1i}, y_{0i}) and (y_{1j}, y_{0j}) have the same conditional distribution given $c_i = c_j$. Hence, $\mathrm{E}(y_{0i} \mid s_i = 1, c_i) = \mathrm{E}(y_{0j} \mid s_j = 0, c_j)$.

Estimate ATT with CIA only

- If CIA holds, we know

$$\mathrm{ATT} = \mathsf{E}(y_i \mid s_i = 1, c_i = c) - \mathsf{E}(y_j \mid s_j = 0, c_j = c).$$

- To estimate ATT, we can use the sample analog of the conditional expectation. This is a matching estimator, of which the details will be discussed latter.
- Advantage: this is "nonparametric". Besides the CIA, we did not impose any other distributional assumption.
- Disadvantage: ATT now is a function of c, hence hard to describe. Matching itself in practice has some issues.

Example of CIA

- Angrist (1998, Estimating the labor market impact on voluntary military service using social security data on military applicants, *Econometrica*).
- the effect of voluntary military service on the later earnings of soldiers.
- $s_i = 1$ if veteran and $= 0$ otherwise.
- He argues that the military screens soldier applicants primarily on the basis of observable covariates such as age, schooling, and test scores.
- and conditional on these observable covariates, veterans and non-veterans are comparable.

Estimate ATT: how does linear regression model help?

- The outcome y_i is generated from

$$y_i = f_i(s_i).$$

  $y_{1i} = f_i(s_i = 1)$ and $y_{0i} = f_i(s_i = 0)$. This is just notational device.

- Assume first that

$$f_i(s_i) = s_i\delta + \eta_i,$$

  and η_i, though might be correlated with s_i, is not a function of s_i.

- δ is the causal (treatment) effect. For discrete $s_i \in \{0, 1\}$,

$$y_{1i} - y_{0i} = f_i(s_i = 1) - f_i(s_i = 0) = 1 \cdot \delta - 0 \cdot \delta = \delta.$$

  This implies that the treatment effect is homogeneous across i. For continuous s_i, $\partial f_i(s)/\partial s$ (usually called partial effect) is the object.

- Next, assume that

$$\eta_i = \mathrm{E}(\eta_i \mid c_i) + (\eta_i - \mathrm{E}(\eta_i \mid c_i)) = c_i'\beta + u_i.$$

  Note $\mathrm{E}(u_i \mid c_i) = 0$.

- Together, we have linear regression model

$$y_i = f_i(s_i) = s_i\delta + c_i'\beta + u_i.$$

- CIA (potential outcomes are conditionally independent of s_i given c_i) here can be rephrased

$$f_i(s) \perp\!\!\!\perp s_i | c_i.$$

  This implies that

$$\mathrm{E}(f_i(s_i = s) \mid s_i, c_i) = \mathrm{E}(f_i(s_i = s) \mid c_i).$$

  It is easy to verify that the above display implies the familiar mean independence assumption $\mathrm{E}(u_i \mid s_i, c_i) = 0$.

- Given $\mathrm{E}(u_i \mid s_i, c_i) = 0$, δ can be consistently estimated by the OLS.
- Conclusion: The mean independence condition $\mathrm{E}(u_i \mid s_i, c_i) = 0$ is a sufficient condition for estimating ATT or ATE (in linear regression model, ATT=ATE) using the OLS.
- Necessary: $\mathrm{Cov}(s_i, u_i) = \mathrm{Cov}(c_i, u_i) = 0$ and $\mathrm{E}(u_i) = 0$.

Limitation of linear regression model

- Linear outcome function $y_i = f_i(s_i) = s_i\delta + c_i'\beta + u_i$.
  - Generalization: discrete choice model, tobit model, semiparametric model (e.g. partially linear model).
- Causal effect is homogeneous across i.
  - Generalization: random coefficient models, finite mixture model.
- Mean independence $\mathrm{E}(u_i \mid s_i, c_i) = 0$ could be violated in some cases.
  - Generalization: panel data model (fixed effect).
- Ad hoc specification about outcome function.
  - Generalization: structural econometric methods.

Basic results about linear regression models

Mechanics of linear regression: setup

- The model is

$$y = x'\beta + u.$$

  We are interested in estimating β, because β can be interpreted as causal effect.

- x can include nonlinear functions of underlying variables, such as log, squares. Mincer's equation,

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u.$$

- Limitation: general nonlinear, e.g. the support of y may be restricted, such as binary; β can be viewed as random β_i.

Data generation process (DGP) and key conditions

- We collect random sample (that is iid) from the underlying population, $\{y_i, x_i : i = 1, \ldots, n\}$, and write

$$y_i = x_i\beta + u_i.$$

Assumption 1 *The key conditions for consistently estimating β are*

$$\mathrm{E}(u_i) = 0, \qquad \mathrm{Cov}(u_i, x_i) = 0.$$

- $\mathrm{E}(u_i) = 0$ is trivial in the presence of intercept. We are usually concerned about $\mathrm{Cov}(u_i, x_i) = 0$.
- $\mathrm{E}(u \mid x) = 0$ is sufficent, which also implies

$$\mathrm{E}(y \mid x) = x'\beta.$$

- Partial effect: $\partial \mathrm{E}(y \mid x)/\partial x_j$ and average partial effect (APE) $\mathrm{E}(\partial \mathrm{E}(y \mid x)/\partial x_j)$.

Three ways of violating $\mathrm{Cov}(u, x) = 0$

- Omitted variables.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u.$$

If x_2 is omitted, the regression becomes

$$y = \beta_0 + \beta_1 x_1 + v, \quad v = \beta_2 x_2 + u.$$

If x_1 and x_2 are correlated, $\mathrm{Cov}(v, x) \neq 0$.

- Measurement error.

$$y = \beta_0 + \beta_1 x_1 + u.$$

Instead of observing x_1, we have only "noisy" observation $x_1^* = x_1 + w$. We have

$$y = \beta_0 + \beta_1 x_1^* - \beta_1 w + u.$$

Then x_1^* containing w is correlated with $u - \beta_1 w$.

- Simultaneity.

$$y = \beta_0 + \beta_1 x_1 + u, \quad x_1 = \alpha_0 + \alpha_1 y + v.$$

x_1 and y can be written as functions of u and v.

OLS estimator

- Let $X = (x_1, x_2, \ldots, x_n)'$ and $Y = (y_1, y_2, \ldots, y_n)'$.
- The OLS estimator is $\hat{\beta} = (X'X)^{-1}(X'Y)$. There are at least two ways to see it.
- First, $\hat{\beta}$ solves $\min_b \sum_{i=1}^n (y_i - x_i' b)^2$ or its FOC $(X'X)b - X'Y = 0$.
- Second, $\hat{\beta}$ is also the sample analog of β, because $\beta = \mathrm{E}(xx')^{-1}\,\mathrm{E}(xy)$. Note that $\hat{\beta} = (n^{-1}X'X)^{-1}(n^{-1}X'Y)$.

- $\beta = \mathsf{E}(xx')^{-1}\,\mathsf{E}(xy)$ follows from

$$xy = xx'\beta + xu \Rightarrow \mathsf{E}(xy) = \mathsf{E}(xx')\beta \Rightarrow \beta = \mathsf{E}(xx')^{-1}\,\mathsf{E}(xy)$$

  Assumption 2 *Clearly, we need to assume* $\mathsf{E}(xx')$ *is nonsingular. This means no perfect multicollinearity.*

- High correlation among regressors (multicollinearity) often cannot be avoided, but not a violation of the assumption. The consequence is larger standard error of $\hat{\beta}$.
- It could be useful to note that $\beta = \mathsf{E}(xx')^{-1}\,\mathsf{E}\big(x\,\mathsf{E}(y \mid x)\big)$. (You do not need to observe y itself).

Asymptotic properties of the OLS estimator

- Consistency relies on replacing Y with $Y = X\beta + U$ in $(X'X)^{-1}(X'Y)$,

$$(X'X)^{-1}(X'(X\beta + U)) = \beta + (n^{-1}X'X)^{-1}n^{-1}X'U.$$

  Note that $n^{-1}X'U \to_p \mathrm{E}(xu) = 0$.

- Let $A = \mathrm{E}(xx')$ and $\hat{A} = n^{-1}X'X$.

- Asymptotic normality follows from

$$\sqrt{n}(\hat{\beta} - \beta) = (\hat{A})^{-1} n^{-1/2} X' U.$$

Note

$$n^{-1/2} X' U = n^{-1/2} \sum_{i=1}^{n} x_i u_i \to_d \mathcal{N}(0, \mathrm{E}(x_i u^2 x_i')),$$

and $\hat{A} \to_p A$.

- Let $B = \mathrm{E}(xx' u^2)$.
- We conclude

$$\sqrt{n}(\hat{\beta} - \beta) \to_d \mathcal{N}(0, A^{-1} B A^{-1}),$$

and

$$\sqrt{n}(\hat{\beta} - \beta) \to_d \mathcal{N}(0, \hat{A}^{-1} B \hat{A}^{-1}).$$

- $B = \mathrm{E}(xx'u^2)$ is unknown.
- Let $\tilde{B} = n^{-1}\sum_{i=1}^{n} x_i x_i' u_i^2$. $\tilde{B} \to_p B$.
- Let $\hat{u}_i = y_i - x_i'\hat{\beta}$ be residuals. Let $\hat{B} = n^{-1}\sum_{i=1}^{n} x_i x_i' \hat{u}_i^2$. It can be shown that

$$\hat{B} - \tilde{B} = o_p(1).$$

  The key step is to write $\hat{u}_i^2 = (x_i(\beta - \hat{\beta}) + u_i)^2$.
- Combining these terms, we have the heteroskedasticity robust variance estimator (a "sandwich") or the so called White estimator.
- Conclusion:

$$\sqrt{n}(\hat{\beta} - \beta) \to_d \mathcal{N}(0, \hat{A}^{-1}\hat{B}\hat{A}^{-1}).$$

Asymptotic normality is useful for statistical inference

- Suppose $\hat{\beta} - \beta \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \hat{A}^{-1}\hat{B}\hat{A}^{-1}$.
- Suppose $\hat{\beta} - \beta$ is a K dimensional vector.
- For any $r \times K$ matrix R, we have

$$R(\hat{\beta} - \beta) \sim \mathcal{N}(0, R\Sigma R').$$

- This fact is used to test the linear restriction about β.

Useful formulas in linear regression model: I

- Linear projection: for any random variable y and a vector x, we can always write

$$y = \theta_0 + x'\theta_1 + u, \quad \theta_1 = \mathrm{Var}(x)^{-1}\,\mathrm{Cov}(x, y) \quad \theta_0 = \mathrm{E}(y) - \mathrm{E}(x)\theta_1.$$

  This also implies $\mathrm{E}(u) = 0$ and $\mathrm{Cov}(x, u) = 0$. Write $L(y|x)$ the linear projection of y on x.

- Let $\theta = (\theta_0, \theta_1')'$ and write $z' = (1, x')$. We can also write

$$\theta = (\mathrm{E}(zz'))^{-1}\,\mathrm{E}(zy).$$

- It can be shown that (θ_0, θ_1) solves the minimum squared error problem,

$$\min_{a,b} \mathrm{E}\Big((y - a - x'b)^2\Big).$$

Useful formulas in linear regression model: II

- Partition linear regression. Consider

$$y = x'\beta_1 + z'\beta_2 + u.$$

We are only interested in β_1. Using linear projection, we can show

$$\beta_1 = [\mathrm{Var}(x - L(x|z))]^{-1}\,\mathrm{Cov}(x - L(x|z), y).$$

$x - L(x|z)$ is the error term of the linear regression of x on z; it is the variation of x by itself.

Useful formulas in linear regression model: III

- Keep using the partition linear regression model formula. The OLS estimator of β_1 is the sample analog of the previous formula,

$$\hat{\beta}_1 = (R_x' R_x)^{-1}(R_x' Y).$$

  Here R_x is the matrix of residuals from regressing x on z. Writing $M_z = I - Z(Z'Z)^{-1}Z'$, $R_x = M_z X$. It is easy to show that

$$\hat{\beta}_1 - \beta_1 = (R_x' R_x)^{-1}(R_x' U).$$

- Under the homoskedasticity assumption, the asymptotic variance of $\hat{\beta}_1$ is

$$\mathrm{E}(u^2)(n\,\mathrm{E}(r_x r_x'))^{-1},$$

  where n is the sample size.

- Larger sample size n decrease variance. Including irrelevant covariates decrease $\mathrm{E}(r_x r_x')$ and increase the variance.

Useful formulas in linear regression model: IV

- If the true model is

$$y = x'\beta_1 + z'\beta_2 + u,$$

and we omit z and run the following regression,

$$y = x'\alpha + v.$$

Then,

$$\alpha = [\mathrm{E}(xx')]^{-1}\,\mathrm{E}(xy) = \beta_1 + \mathrm{E}(xx')^{-1}\,\mathrm{E}(xz')\beta_2.$$

- Unless x and z are uncorrelated or unless $\beta_2 = 0$, omitted variables create bias.

Useful formulas in linear regress model: V

- SST (total sum of squares), SSE (explained sum of squares) and SSR (sum of squared residuals).

$$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2, \quad SSE = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2, \quad SSR = \sum_{i=1}^{n}\hat{u}_i^2.$$

  Here $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$ and $\hat{u}_i = y_i - \hat{y}_i$.
- SST = SSE + SSR, because $\sum_{i=1}^{n}\hat{u}_i(\hat{y}_i - \bar{y}) = 0$.
- $\sum_{i=1}^{n}\hat{u}_i(\hat{y}_i - \bar{y}) = 0$ follows from the FOC $\sum_{i=1}^{n}(y_i - x_i\hat{\beta})x_i = 0$.
- $R^2 = 1 - SSR/SST$.

Practical concerns

- Do not always attempt to increase R^2, adjusted R^2, or some other goodness-of-fit measure. You can decrease the standard error of β estimates.
- Mathematically, R^2 is not informative. View $\hat{y}_i$ as your forecast for y_i, R^2 is not a good measure.

$y_1 = 1$	$y_2 = -1$	$y_3 = 1$	$y_4 = -1$
$\hat{y}_1 = 0.8$	$\hat{y}_2 = -0.8$	$\hat{y}_3 = 0.8$	$\hat{y}_4 = -0.8$
$\tilde{y}_1 = -1$	$\tilde{y}_2 = 1$	$\tilde{y}_3 = -1$	$\tilde{y}_4 = 1$

Then $\bar{y} = 0$. The R^2 from $\hat{y}_i$ is 0.64. However, the bad "forecast" from $\tilde{y}_i$ has $R^2 = 1$.

- Include covariates that help predict the outcome if they are uncorrelated with the covariates of interest.
- The asymptotic variance of $\hat{\beta}_1$ is

$$\mathrm{E}(u^2)(n\,\mathrm{E}(r_x r_x'))^{-1},$$

- One control that can substantially reduce the error variance is a lagged value of the outcome y.

Instrumental variables regression

Single instrumental variable

- Consider

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + u, \quad \mathrm{E}(u) = 0, \mathrm{Cov}(x_j, u) = 0, j = 1, \ldots, K-1,$$

  but $\mathrm{Cov}(x_K, u) \neq 0$.
- An IV z_1 satisfies (i) $\mathrm{Cov}(z_1, u) = 0$ (exogeneity); (ii) z_1 is partially correlated with x_K *after taking account of* the other exogenous variables (relevance).
- Condition (ii) means in the linear projection of x_K onto all the exogenous variables:

$$x_K = \delta_0 + \delta_1 x_1 + \cdots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_k,$$

  $\theta_1 \neq 0$.

- Let $z = (1, x_1, \ldots, x_{K-1}, z_1)$ be the vector of all exogenous variables. We have $\mathrm{E}(uz) = 0$, and

$$\mathrm{E}(zy) = \mathrm{E}(zx')\beta + \mathrm{E}(zu) = \mathrm{E}(zx')\beta + 0.$$

  So long as $\mathrm{E}(zx')$ has full rank, we have

$$\beta = \mathrm{E}(zx')^{-1}\,\mathrm{E}(zy)$$

  Then the IV estimator is

$$\hat{\beta}_{iv} = (\sum_{i=1}^{n} z_i x_i')^{-1} \sum_{i=1}^{n} z_i y_i$$

- To ensure the nonsingularity of $\mathrm{E}(zx')$, we must have $\delta_K \neq 0$.

Probability limit

- A "slightly" endogenous IV may produce worse estimate than OLS: consider

$$y = \beta_0 + \beta_1 x + u,$$

where x is a scalar and $\mathrm{E}(xu) \neq 0$.

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\sigma_u}{\sigma_x} . \operatorname{Corr}(x, u).$$

and

$$\operatorname{plim} \hat{\beta}_{1,iv} = \beta_1 + \frac{\sigma_u}{\sigma_x} \cdot \frac{\operatorname{Corr}(z, u)}{\operatorname{Corr}(z, x)}.$$

- If $\operatorname{Corr}(z, x)$ is small (weak IV), then even small $\operatorname{Corr}(z, u)$ lead to large bias.

General case

- In some cases, we have more instruments than we need.

$$y = x'\beta + u, \quad \mathrm{E}(zu) = 0,$$

  where $L = \dim(z) \geq \dim(x) = K$.
- Example: let $x = (1, x_1, x_2, x_3)'$ with x_2 and x_3 being endogenous. Suppose there are two instruments z_1 and z_2 for x_2 and one instrument z_3 for z_3. Then the vector of all exogenous variables $z = (1, x_1, z_1, z_2, z_3)$. Here $L = 5$ and $K = 4$.
- When $L > K$, we say the model is (potentially) *overidentified*.
- When $L = K$, the model is *just identified*.

- Suppose z_1 and z_2 are IVs for x_K. Any linear combination of z_1 and z_2 can be a valid IV.
- Which should we use? Under a homoskedasticity assumption, the best IV for x_K is the linear projection of x_K on *all* exogenous variables z.

- We can write

$$x = \Pi' z + r$$

where Π is the $L \times K$ matrix

$$\Pi = \mathrm{E}(zz')^{-1}\,\mathrm{E}(zx') \quad \text{and} \quad \mathrm{E}(zr') = 0.$$

Note that

$$zx' = \begin{bmatrix} zx_1 & zx_2 & \dots & zx_K \end{bmatrix}.$$

The above formula is identical to do the linear projection of each x_j onto z and stack them together by column.

- For each x_j we can write

$$x_j = z\pi_j + r_j \equiv x_j^* + r_j$$

where π_j is the jth column of Π.

- For any $x_j \in z$, $x_j^* = x_j$, so exogenous variables act as their own instruments.

- In the general case, use

$$x^* = \Pi' z.$$

  as the $K \times 1$ vector of instruments for x.

- Because z is exogenous, so is x^*: $\mathrm{E}(x^* u) = 0$.
- We have $\beta = \mathrm{E}(x^* x)^{-1}\,\mathrm{E}(x^* y)$. For the estimation, we just need an estimator $\hat{x}^*$ of x^*:

$$\hat{X}^* = Z\hat{\Pi}, \quad \hat{\Pi} = (Z'Z)^{-1}(Z'X)$$

$$\hat{\beta} = \underbrace{(\hat{X}^{*\prime} X)^{-1}(\hat{X}^{*\prime} Y)}_{\text{IV estimator}} = \underbrace{(\hat{X}^{*\prime} \hat{X}^*)^{-1}(\hat{X}^{*\prime} Y)}_{\text{2SLS estimator}}.$$

- Do not run 2SLS manually. The reported variance in the second stage is misleading.

Asymptotic normality of 2SLS IV estimator

- Write $P_z = Z(Z'Z)^{-1}Z'$.
- It is useful to write

$$\hat{\beta}_{2sls} - \beta = (n^{-1}X'P_zX)^{-1}(n^{-1}X'P_zU).$$

- The consistency follows from $n^{-1}X'P_zU \to_p 0$.
- The asymptotic normality follows from

$$\sqrt{n}(\hat{\beta}_{2sls} - \beta) \to_d \mathcal{N}(0, A^{-1}BA^{-1}),$$

where

$$A = \mathrm{E}\left(x^* x^{*'}\right), \quad B = \mathrm{E}\left(u_i^2 x^* x^*\right).$$

$$\hat{A} = n^{-1}X'P_zX \to_p A \quad \hat{B} = n^{-1}X'P_z\hat{U} \to_p B.$$